

Zeel Gajjar

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PROFESSIONAL SUMMARY

Aspiring Cybersecurity researcher and Quantum Computing enthusiast with a strong foundation in cryptographic systems and security research. Demonstrated expertise in developing advanced cryptographic tools and conducting noise-resilient quantum simulation research. National-level speaker and award-winning innovator recognized for bridging the gap between classical security and post-quantum transitions. Active IEEE leader and CTF competitor focused on solving complex security challenges.

EDUCATION

M.B. Patel Institute of Technology, CVM University

Gujarat, India

B.Tech in Cybersecurity

Expected 2029

- **Relevant Coursework:** Cryptography, Network Security, Quantum Computing, Signal Processing, Algebraic Topology, Binary/Web Exploitation, Higher Order Linear ODEs.

EXPERIENCE

Research Intern — QKD & RSA

Jun 2026 — Present

TechForGood 2026, IEEE MIT-ADT Student Branch

Remote

- Researching Quantum Key Distribution (QKD) protocols and RSA systems under IEEE Maharashtra Section.
- Investigating transitions to post-quantum standards, including lattice-based and hash-based schemes.

RESEARCH & PRESENTATIONS

DAU Silver Jubilee Seminar (ETSIF), Decoherence Effects in VQE of ϕ^4 Scalar Field Theory

2026

- Implemented VQE for 4-site lattice ϕ^4 Hamiltonian using Qiskit; achieved 0.87% accuracy vs. exact diagonalization.
- Characterized noise effects (depolarizing, damping) and demonstrated robustness across multiple ansatz depths.

PDEU — Indo Quantum Summit 2026, Quantum-Inspired Random Number Generator

2026

- Designed entropy harvesting combining photon shot noise and thermal noise; achieved 98.3% NIST SP 800-22 pass rate.

SELECTED PROJECTS

Quantum-Inspired Behavioral Authentication

2026 — Ongoing

- Designing continuous authentication encoding dynamics as quantum state vectors; implementing CRYSTALS-Kyber.

Cryptographic Key Statistical Test Suite

Feb 2026

- Implemented 34+ tests (NIST SP 800-22 battery + advanced tests) using Python/NumPy/SciPy.

LEADERSHIP & ACHIEVEMENTS

- **Vice Chairperson:** IEEE Signal Processing Society, M.B. Patel Institute of Technology.
- **1st Place:** State-Level Ideathon (AI-integrated firewall with cryptographic anomaly detection).
- **1st Place:** College-Level Ideathon (IoT encryption system using camera-based entropy extraction).
- **CTF Competitor:** Active in CryptoHack and OverTheWire (Web, Binary, Cryptography).

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Julia, Bash, HTML/CSS.
- **Cryptography:** NIST testing, QKD (BB84), RSA, Post-Quantum (Kyber, SPHINCS+), Cryptanalysis.
- **Quantum & Signal:** Qiskit, VQE, NISQ noise simulation, FFT/STFT, EEG analysis.
- **Security Tools:** Burp Suite, Wireshark, Nmap, OpenCV, Vercel.